Arjun Nanduri

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Education

University of California: Berkeley, BA in Computer Science — Expected May 2028

- GPA: 3.9 | SAT: 1590
- Courses: Probability & Stochastic Processes, Machine Learning, Computer Security, Data Structures

Experience

Machine Learning Engineer, Aver Technologies Inc. – Virginia — Aug 2025 – Present

- Building a multi-stage PyTorch CV pipeline to automate depth tracking when driving piles (foundational support for heavy structures, like bridges) into the ground — **replacing a manual, hand-marked DOT compliance process**.
- Pipeline processes **1000fps** pile-driving footage — YOLO detection and segmentation to track painted depth markings, audio-visual sync to isolate hammer strikes, and depth-map pixel-to-real-world conversion.
- Conducted field interviews with construction workers to surface real-world constraints — inconsistent hand-drawn markings, Spanish-speaking workforce — **directly informing model design decisions**.

Software Engineering Intern, CACI International Inc. – Virginia — May 2025 – Aug 2025

- Built a full-stack logistics web app for **U.S. Customs field officers** using Angular, Spring Boot, and PostgreSQL.
- Expanded database test coverage **by 20%**, enhancing system stability and enabling more efficient bug detection.
- Ramped onto a large, unfamiliar Spring Boot codebase fast enough to **pick up frontend bug tickets and a backend bug-fix story**, contributing through sprint planning, standups, and code reviews in an Agile team.

Data Science Research Intern, CDSS, UC Berkeley – Berkeley, CA — Jan 2025 – May 2025

- Fine-tuned BART transformer for Ancient Akkadian lemmatization in low-resource settings, **beating open-source baselines by 15%** using a custom edit-distance metric over lemma forms.
- Built preprocessing and alignment pipelines for structured linguistic data from ORACC's cuneiform corpus (∼**1M annotated words across** ∼**10K texts**).
- Stepped up to lead disorganized team with no formal authority — **built shared task system**, assigned work by strength, held mock presentations, presented at UC Berkeley Discovery Symposium.

Projects

PotBot — Civic Reporting Agent, *CalHacks Prize Winner* — https://devpost.com/software/potbot

- Won sponsor prize at CalHacks — led team of 3, owned backend and CV pipeline, building an agentic pipeline with a Llama vision model to validate infrastructure damage photos and auto-submit repair requests to SF city government via Playwright; **shipped working SF demo in 36 hours** over an unfinished feature set.

DealScout — Resale Arbitrage Agent (LangGraph, Playwright) — github.com/FavouritePlayer/DealScout

- Built a LangGraph-orchestrated agent that scrapes live Craigslist listings via Playwright, scores resale value with deterministic arithmetic kept separate from LLM judgment, then restricted image-attachment to post-filter candidates to **avoid a ∼4x blowup in scan time**.

Context Custodian — Workspace Hygiene Agent (FastAPI) — github.com/FavouritePlayer/ContextCustodian

- Built a FastAPI agent that compacts and cleans an agent-first workspace for downstream agents to ingest — a three-stage prompt-injection detection pipeline (regex, vector similarity, LLM adjudication) with a fail-open fallback, where **every fix is a reversible move, never a delete**.

Quadruped Locomotion with PPO (MuJoCo, Gymnasium, SB3) — github.com/FavouritePlayer/ant_sim

- Trained and validated quadruped locomotion policies (PPO, MuJoCo) through an **automated regression-test suite and multi-seed replication**, shipping a policy with 2.1x the flat-baseline reward on unseen terrain.

Face Morpher — From-Scratch Warping (NumPy, dlib, scipy) — github.com/FavouritePlayer/face-morpher

- Replaced OpenCV's warp with a **custom NumPy warp** (affine solve, inverse mapping, bilinear interpolation).
- Solved a per-triangle affine transform from 3-point landmark correspondences (**3-point** $\rightarrow$ **2**$\times$**3 matrix**), combined with polygon-masked, piecewise Delaunay-triangle warping to produce image morphs and mean faces.

Skills

Languages: Python, Java, JavaScript, SQL, Dart, Scheme

Technologies: React.js, Angular, Spring Boot, Flutter, Next.js, Firebase, MongoDB, Git, PyTorch, scikit-learn, Pandas, LangGraph, FastAPI, Playwright, MuJoCo, Gymnasium, Stable-Baselines3.